



HIMACHAL PRADESH TECHNICAL UNIVERSITY
L.R Institute of Engineering & Technology, Solan | Sortiq Solutions Pvt. Ltd.



CAMPUS NOTES EXCHANGE PLATFORM

A Gamified Peer-to-Peer Academic Resource Sharing & Collaboration System

React 19 & Vite

Node.js & Express

MongoDB & Mongoose

Cloudinary CDN

Capacitor Mobile

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The Problem: Academic Resource Inefficiency on Campuses

1. Fragmented Channels

- Study notes scattered across ephemeral WhatsApp groups, Telegram channels, and drives.
- Crucial exam notes shared in mid-semester become impossible to find during finals.
- Zero centralized institutional catalog or curriculum structure.

2. No Quality Control

- Notes are circulated with no syllabus verification or subject code matching.
- Students frequently waste revision time studying outdated or erroneous materials.
- No community rating or validation to identify high-quality notes.

3. Absence of Incentive

- High-performing students invest hours making handwritten notes but have no motivation to share.
- Traditional sharing is one-sided with zero peer recognition or reward loop.
- Leads to a culture of resource hoarding before exams.

4. Storage & Security Risks

- Free cloud drive links expire, reach download limits, or break when files move.
- Direct peer downloads on unvetted channels risk circulating malware (.exe, scripts).
- Large PDFs clog student smartphone storage.

The Solution: A Unified, Gamified Academic Hub

Centralized Taxonomy

- Organized repository classified by Branch, Semester (1-8), Subject, and Unit.
- Instant discovery through multi-field keyword regular expression search.
- Eliminates reliance on fragmented messaging apps and broken drive links.

Credit Gamification

- Incentivizes peer contributions through an automated credit reward economy.
- Uploaders earn +10 credits on upload and +5 passive credits upon peer downloads.
- Campus-wide leaderboard highlights top academic contributors.

Community Trust Layer

- Embedded 1-5 star peer rating system with automatic average recalculation.
- Threaded discussion comments for peer clarification and feedback.
- Student reporting system for copyright and syllabus moderation.

Hybrid Availability

- Cloudinary remote CDN for cloud storage with automatic local disk fallback.
- Zero-cost deployment over campus Wi-Fi local area network (LAN).
- Capacitor integration for native Android mobile accessibility.

High-Level Three-Tier Architecture & Data Flow

1. PRESENTATION TIER (Client)

- React 19 Single Page Application (SPA) powered by Vite.
- Tailwind CSS for responsive design across mobile and desktop breakpoints.
- Framer Motion for micro-animations and Canvas Confetti feedback.
- Axios HTTP client with automatic JWT Bearer token request interceptors.
- Capacitor Android Container: Hosts compiled React bundle inside native WebView container for offline/LAN campus app.

2. APPLICATION TIER (Server)

- Node.js & Express.js RESTful API handling business logic and routing.
- JWT Verification & Real-time Banishment Guard (authMiddleware.js).
- Role-Based Access Control (RBAC) protecting administrative actions.
- Multer File Streaming Pipeline with strict MIME whitelist (25MB limit).
- Gamification & Rating Calculation Hooks running on MongoDB events.

3. DATA & STORAGE TIER

- MongoDB NoSQL document database accessed via Mongoose ODM.
- Embedded Subdocuments: Ratings and comments stored inside Note schema for zero-join sub-50ms reads.
- Cloudinary Remote CDN: Secure streaming of PDF and image assets.
- Local Storage Fallback: Writes to /server/uploads for isolated campus Wi-Fi networks with zero cloud dependencies.

Technology Stack & Engineering Rationale

Frontend Technologies

- React 19 & Vite: Fast Virtual DOM diffing with sub-second HMR updates.
- Tailwind CSS: Utility-first CSS ensuring lightweight, responsive layouts.
- Lucide React: Scalable vector iconography for modern UI aesthetics.

Backend Technologies

- Node.js & Express.js: Event-driven, non-blocking asynchronous REST APIs.
- Multer: Streaming buffer processing for high-volume multipart form uploads.
- CORS: Controlled cross-origin access between frontend and backend ports.

Database & Cloud Storage

- MongoDB & Mongoose: Flexible document schema eliminates multi-table SQL joins.
- Cloudinary CDN: Remote cloud storage with fast CDN delivery and preview URLs.
- Local Fallback: Auto-serves from Express static path if offline.

Security & Mobile Runtimes

- Bcryptjs (10 Rounds): Cryptographically salted one-way password encryption.
- JSON Web Tokens (JWT): Stateless bearer tokens with payload verification.
- Capacitor: Builds standalone native Android APK from single web codebase.

Database Design: Schemas & Embedded Subdocuments

User Schema (User.js)

- `_id`: ObjectId (Primary Key)
- name, email (Unique), rollNumber
- password: 60-char Bcrypt salted hash
- branch, semester: Syllabus mapping
- credits: Accumulated gamification points
- bookmarks: Array of referenced Note IDs
- role: 'user' | 'admin' (RBAC)
- isBanned: Real-time suspension flag

Note Schema (Note.js)

- `_id`: ObjectId (Primary Key)
- title, subject, branch, semester, unit
- teacherName, tags []: Search keywords
- fileUrl: Cloudinary or local upload path
- uploadedBy: Ref to User (Author)
- views, downloads: Metric counters
- averageRating: Pre-calculated score
- EMBEDDED SUBDOCUMENTS:
 - ratings [{ user, score: 1-5 }]
 - comments [{ user, userName, text }]

Performance Strategy

- Why MongoDB Embedded Subdocuments?
 - Ratings & comments are only read when viewing the parent note.
 - Storing them inside Note.js eliminates costly SQL JOIN operations.
 - Delivers sub-50ms single-query document retrieval.
- Report Schema (Report.js):
 - Stores reportedUser, reportedBy, reason, and moderation status for admin review.

Student Experience: Core Functional Modules

Smart Explore & Search

- Multi-criteria filtering by Branch, Semester, and Subject code.
- Regex search across note titles, descriptions, teacher names, and tags.
- Real-time debounced updates without full page refreshes.

Note Preview & Details

- Interactive modal popup with instant PDF/image viewing.
- Displays uploader profile, download counter, and view statistics.
- Direct secure download link and one-click bookmarking.

Peer Rating & Feedback

- In-place 1 to 5 star rating submission.
- Mongoose pre-save hook automatically updates the arithmetic average.
- Threaded discussion comments for peer Q&A and exam tips.

Gamified Upload Suite

- Drag-and-drop file upload with client & server validation.
- Syllabus unit and instructor tagging fields.
- Celebratory Canvas Confetti animation upon successful upload.

Credit Economy & Gamification Flywheel

Step 1: Upload (+10)

- Student scans and uploads high-quality notes or PPT slides.
- System allocates +10 credit points to the student's profile instantly.
- Confetti micro-animation delivers immediate psychological gratification.

Step 2: Peer Discovery

- Peers discover the notes via branch/semester filters before exams.
- Quality preview lets peers inspect syllabus match before downloading.
- Transparent star ratings guide peers to the most reliable notes.

Step 3: Royalty (+5)

- Whenever another student downloads the note, the author receives +5 credits.
- Creates a passive dividend loop: better notes yield higher recurring credits.
- Prevents self-download credit farming via author identity checks.

Step 4: Leaderboard

- Institute-wide leaderboard ranks top student contributors.
- Fosters healthy academic recognition across departments.
- Transforms note-sharing from a chore into an engaging campus activity.

Enterprise Security & Content Moderation

Bcrypt Password Hashing

- Passwords never stored in plaintext.
- Mongoose pre-save hook generates 10-round cryptographic salt.
- Protects against dictionary and rainbow-table attacks.

Stateless JWT Auth

- Signed HMAC SHA-256 tokens authenticate client requests.
- Axios interceptor injects Bearer token into outgoing headers.
- Query parameter token support enables direct authenticated downloads.

Multifile Sanitization

- Strict extension whitelist: .pdf, .ppt, .pptx, .jpg, .jpeg, .png.
- Executable extensions (.exe, .bat, .sh) rejected before buffer processing.
- Strict 25MB file size limit prevents Denial of Service (DoS).

Instant Banishment & RBAC

- Middleware checks user.isBanned flag on every protected request.
- Banned accounts immediately receive HTTP 403 Forbidden.
- Admin console facilitates one-click user ban and content cleanup.

Mobile Readiness & Campus Wi-Fi Networking

Capacitor Mobile Build

- Wraps React production build (dist/) in native Android container.
- Generates installable .apk without rebuilding frontend in React Native.
- 100% web code reuse with native mobile performance.

Zero-Cost Campus LAN

- Node.js backend binds to host machine's Wi-Fi network interface IP.
- Students access platform over college Wi-Fi (e.g., 192.168.1.45:5000).
- Requires zero external server fees or internet dependency.

Dynamic Host Resolution

- Client uses `window.location.hostname` to dynamically target backend API.
- Prevents static IP mismatch errors when moving across campus Wi-Fi APs.
- Smooth transition between development and staging environments.

Cloud Production Path

- Tested deployment roadmap for Render / Railway cloud servers.
- MongoDB Atlas M0 Free Tier for resilient cloud database hosting.
- Cloudinary CDN ensures persistent document storage across restarts.

Quality Assurance: Testing & Performance Benchmarks

Comprehensive Test Matrix

- TC-01 to TC-04: User registration & JWT authentication passed.
- TC-05 to TC-07: Valid PDF upload stored; malicious executables (.exe) and oversized files (>25MB) successfully blocked.
- TC-08 & TC-09: Regex keyword search & multi-select branch filtering verified.
- TC-10 to TC-12: Download counter + author credit increment (+5) validated.
- TC-13 & TC-14: Bookmarking & instant user banishment (HTTP 403) verified.
- TC-15 & TC-16: Local campus Wi-Fi access & disk fallback validated.
- Result: 16 / 16 Test Cases Passed (100% Pass Rate).

Performance Benchmarks

- Sub-50ms Read Queries: Embedded subdocuments eliminate SQL joins, delivering instantaneous note view and comment loading.
- Lightweight Bundle: Vite tree-shaking yields optimized production JavaScript bundles for fast initial page load.
- Hybrid Resilience: Server automatically falls back to local disk storage if Cloudinary CDN credentials are unset or offline.
- Responsive Breakpoints: Validated on mobile (360px), tablet (768px), and desktop (1080p) screens with zero layout distortion.
- Zero Memory Leaks: Multer memory buffers cleanly cleared after Cloudinary upload completes.

Conclusion & Future Roadmap

Project Summary

- Delivered a production-ready, peer-to-peer academic resource ecosystem.
- Solves campus resource fragmentation by centralizing notes by branch, semester, and unit.
- Incentivizes active student participation through an organic credit-based gamification loop.
- Robust enterprise security with Bcrypt, stateless JWT, and instant ban enforcement.
- Successfully developed and tested during industrial internship at Sortiq Solutions Pvt. Ltd.

Future Roadmap

- AI Note Summarizer: Automated bulleted revision summaries & key formula generation from uploaded PDFs using LLMs.
- OCR Handwritten Search: Optical Character Recognition (Tesseract) to search text inside handwritten notes.
- Push Notifications: Real-time alerts when new notes are uploaded for enrolled courses.
- Campus Perk Store: Redeem earned gamification credits for campus bookshop and printing discounts.
- THANK YOU! Open for Viva Examination & Demo Questions.